MILITARY SPECIFICATION

CIRCUIT CARD ASSEMBLY - DEMODULATOR

This specification is approved for use by the Mobility Equipment Research and Development Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a demodulator used in the data receiver, R-1861/FSS-9(V), a component of the Joint-Services-Interior Intrusion Detection System (J-SIIDS). The demodulator consists of a single circuit card assembly.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ-S-781	- Strapping, Steel, and Seals.
PPP-B-566	- Boxes, Folding, Paper Board.
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-636	- Boxes, Shipping, Fiberboard.
PPP-B-676	- Boxes, Setup.

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MIL-P-116	- Preservation-Packaging, Methods of.
MIL-T-52877	- Transmission Group Data: Transmitter, Data: T-1257/FSS-9(V) and Receiver, Data: R-1861/FSS-9(V).

FSC 6350

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Mobility Equipment Research and Development Command, ATTN: DRDME-DS, Fort Belvoir, VA 22060 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

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MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-454	- Standard General Requirements for Electronic Equipment.
MIL-STD-781	- Reliability Design Qualification and Production Acceptance Tests: Exponential Distribution.
MIL-STD-810	- Environmental Test Methods.
MIL-STD-1188	- Commercial Packaging of Supplies and Equipment.
MIL-STD-45662	- Calibration System Requirements.

DRAWINGS

ME

TA13220E3550	- Printed Wiring Board Assembly, A4 (Channel 2).
D13220E3604	- Data transmitter A-3 PWB.

(Copies of specifications, standards, and drawings required by the contractor in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Description. The demodulator shall be in accordance with Top Assembly TA13220E3550, and as specified herein.

3.1.1 Function. The demodulator is an integral part of the data receiver, R-1861/FSS-9(V), which is a major component of the data transmillion group conforming to MIL-T-52877. The demodulator detects the frequency shift keyed (FSK) signal received from the data transmitter and converts this FSK signal into a digital data stream for processing by other digital circuits located in the data receiver.

3.1.2 Drawings. The drawings forming a part of this specification are end product drawings. No deviation from the prescribed dimensions or tolerances is permissible without prior approval of the contracting officer. Where tolerances could cumulatively result in incorrect fits, the contractor shall provide tolerances within those prescribed on the drawings to insure correct fit, assembly, interchangeability and operation of the demodulator. Any data (e.g., shop drawing, layouts, flow sheets, processing procedures, etc.) prepared by the contractor or obtained from a vendor to support fabrication and manufacture of the production item shall be made available, upon request, for inspection by the contracting officer or his designated representative.

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3.2 First article (preproduction model). The contractor shall furnish one or more demodulators as specified (see 6.2) for examination and testing within the time frame specified (see 6.2) to prove prior to starting production that his production methods will produce demodulators that comply with the requirements of this specification. Examination and tests shall be as specified in Section 4; and, unless otherwise specified herein, all examinations and tests shall be conducted by the contractor subject to surveillance and approval by the Government (see 6.3). When specified (see 6.2), the Government will conduct any or all of the preproduction examination and tests, as specified (see 6.2).

3.3 Material. Material shall be as specified herein and on the applicable drawings. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification (see 6.5).

3.4 Environmental conditions. Each demodulator shall withstand individual environmental conditions specified herein, or combinations thereof, without damage or malfunction when tested as specified in 4.5.2.5.

3.4.1 Operational.

(a) Temperature range of -20° F to +150° F.
(b) Humidity up to 95 percent relative.

3.4.2 Nonoperational and storage.

(a) Temperature range: - 30 F to +165° F.
(b) Humidity: up to 95 percent relative.

3.5 Detailed technical requirements.

3.5.1 Input signal. The demodulator shall be capable of demodulating the continuous, 12 Hertz (Hz) frequency shift keyed (FSK) data signal provided by the Data Transmitter, T-1257/FSS-9(V). The two FSK tones utilized in the received data signal are 1070 Hz and 1270 Hz.

3.5.2 Power requirements. The demodulator shall operate on 20 ± 1 Volt direct current (Vdc) and the maximum current drain shall not exceed 5 milliamperes (mA).

3.5.3 Voltage output. The demodulator voltage output at TP4 (reference TP7), shall be 6.8 ± 0.2 Vdc.

3.5.4 Bandpass. The demodulator bandpass filters shall be centered at 1070 ± 5 Hz and 1270 ± 5 Hz respectively. The 3 decibel (dB) bandpass of the 1070 Hz filter (TP1 reference TP7) shall be 70 ± 10 Hz. The 3 dB bandpass of the 1270 Hz filter (TP2 reference TP7) shall be 70 ± 10 Hz.

3.5.5 Sensitivity. The base band data (TP8 reference TP7) shall not drop out until the signal level at the input is less than -33 ± 2 decibels referred to one milliwatt (dBm). Sensitivity of the 1070 Hz and 1270 Hz discriminator circuits shall be within 1 dB of each other.

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3.5.6 Line fault sensitivity. The line fault output (TP3 reference TP7) shall change state, high (18 ± 1 Vdc) to low (0 to 1 Vdc), when the signal level at the input is + 0.5 ± 0.5 dBm greater than the level at which base band dropout occurs.

3.6 Government-loaned property. Unless otherwise specified (see 6.2), the following property in the quantity indicated will be loaned by the Government (see 6.4), for testing the demodulators.

Item No.	Description	Identification	Quantity
1.	Data Transmitter A3 PWB	D13220E3604-1	1

3.7 Interchangeability. All parts having the same part number shall be funcionally and dimensionally interchangeable. Interchangeable parts are defined as two or more like parts possessing such functional and physical characteristics as to be equivalent in performance and durability, and capable of being exchanged one for the other without alteration of the parts themselves or of adjoining parts, except for adjustment, and without selection for fit or performance.

3.8 Reliability. The demodulator shall demonstrate a Mean-Time-Between-Failure (MTBF) of 10,000 hours or greater when tested as specified in 4.5.2.6.

3.9 Workmanship. Workmanship shall be in accordance with MIL-STD-454, Requirement 9.

4 QUALITY ASSURENCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapporved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. The contractor is responsible for insuring that components and materials used are manufactured, examined and tested in accordance with referenced specifications, standards and drawings, as applicable.

4.1.2 Disassembly inspection. Failure of any test or examination by the preproduction model shall be cause for disassembly, in the presence of a Government representative, of the preproduction model to the extent necessary to determine the cause of the failure. Each disassembled part shall be examined in detail for compliance with this specification and referenced drawings in regard

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to dimensions, tolerances, and workmanship. Parts not complying with such requirements shall be rejected and shall be cause for rejection of the preproduction model. Reassembly with replacement parts and retesting shall be the responsibility of the contractor.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) Preproduction inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).
(c) Inspection of packaging (see 4.6).

4.3 Preproduction inspection.

4.3.1 Examination. The preproduction demodulator shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection or for performing the disassembly inspection specified in 4.1.2.

4.3.2 Tests. The preproduction demodulator shall be subjected to the tests marked "X" in column 1 of table I. Failure of any test shall be cause for rejection or for performing the disassembly inspection specified in 4.1.2.

4.4 Quality conformance inspection.

4.4.1 Sampling. Sampling for examination shall be in accordance with MIL-STD-105, inspection level II. Sampling for tests shall be in accordance with MIL-STD-105, inspection level S-2.

4.4.2 Examination. Samples selected in accordance with 4.4.1 shall be examined for the defects specified in 4.5.1. AQL shall be 2.5 percent defective.

4.4.3 Tests.

4.4.3.1 Individual. Each demodulator, including those selected for sample test, shall be subjected to the tests marked "X" in column 2 of table I. Failure of any test shall be cause for rejection.

4.4.3.2 Samples. Samples selected in accordance with 4.4.1 shall be subjected to the tests marked "X" in column 3 of table I. AQL shall be 2.5 percent defective.

4.5 Inspection procedure.

4.5.1 Examination. The demodulator shall be examined as specified herein for the following defects:

Major

101. Any part (or component) not in accordance with the drawings.
102. Components missing or not as specified.

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103. Material not as specified.
104. Interchangeability not as specified.
105. Workmanship not as specified.

4.5.2 Tests.

4.5.2.1 Test condition. Unless otherwise specified herein, tests shall be conducted at prevailing ambient temperature and humidity conditions. Prior to testing, each demodulator shall be aligned in accordance with the alignment procedure in the appendix. After alignment each demodulator shall be subjected to a 72-hour burn-in during which it shall be exercised through the appendix test method paragraph 4(a) and (f) at a rate of twice during each 24 hour period (once in the morning and once in the afternoon). All failure shall be recorded and analyzed for cause of failure. Corrective action shall be taken when appropriate. Copies of these failure reports shall be forwarded to the contracting officer. Units for testing shall be selected from those units having successfully completed the 72-hour burn-in.

4.5.2.2 Test schedule.

TABLE I. Demodulator test schedule.

Preproduction	Quality Conformance		Test	Requirement Paragraph	Test Paragraph
	Individual	Sample			
1	2	3	4	5	6
X	X		Test condition (Burn-in)		4.5.2.1
X	X	-	Functional	3.5	4.5.2.4
X	-	-	Reliability	3.8	4.5.2.6
X	-	X	Low temperature	3.4.1 (a), 3.4.2 (a)	4.5.2.5.2
X	-	X	High temperature	3.4.1 (a), 3.4.2 (a),	4.5.2.5.3
X	-	X	Humidity	3.4.1 (b), 3.4.2 (b)	4.5.2.5.4

4.5.2.3 Test methods. Tests shall be conducted in accordance with the appendix test method and as specified herein. Calibration of test equipment shall be in accordance with MIL-STD-45662.

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4.5.2.4 Functional. The demodulator shall be tested in accordance with the appendix test method. Nonconformance to 3.5 shall constitute failure of this test.

4.5.2.5 Environmental. The demodulator shall be subjected to the following series of tests specified in MIL-STD-810.

4.5.2.5.1 Functional verification. Functional verification testing required for environmental testing shall be conducted in accordance with the appendix test method paragraph 4(d) and (f).

4.5.2.5.2 Low temperature. Low temperature test method 502.1, procedure I, storage temperature -30° F for not less than 48 hours, equipment operating temperature -20° F.

4.5.2.5.3 High temperature. High temperature test method 501.1, procedure II. Except that the storage temperature shall be +165° F; the operating temperature shall be +150° F.

4.5.2.5.4 Humidity. Method 507.1, procedure I with the following exceptions:

(a) Step 5 shall be for 5 cycles (not less than 120 hours). Measurements at the completion of each cycle.
(b) During steps 2 and 3, the relative humidity shall be maintained at 95 percent or greater.
(c) During step 4, the relative humidity shall be maintained at 92 percent or greater.
(d) Measurement shall commence 1 hour and 15 minutes prior to the completion of each 24-hour cycle.

The demodulator may be removed from the environmental chamber for measurement; however, unsuccessful completion of measurements or unsuccessful return of the demodulator to the test chamber within 1 hour and 15 minutes subsequent to the initiation of measurements shall constitute failure of the test.

4.5.2.5.5 Failure criteria. Nonconformance to 3.4 shall constitute failure of these tests.

4.5.2.6 Reliability. Using the upper MTBF (θ_0) of 10,000 hours and a lower test MTBF (θ_1) of 5000 hours, the demodulator shall be tested as specified herein with "accept" and "reject" criteria in accordance with MIL-STD-781, test plan VIIIC. Continue testing until either an "accept" or "reject" decision is reached. A failure is defined as any malfunction which causes or may cause one or more of the following:

(a) Failure to commence operation, cessation of operation, or degradation of performance below the designated levels.
(b) Serious damage to the system.
(c) Serious personnel safety hazards.

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4.5.2.6.1 Test procedure. The testing shall be conducted under ambient laboratory conditions, without temperature or humidity cycling. Operation shall be continuous throughout the test. The demodulator shall be exercised through the appendix test method paragraph 4(d) and (f) at least twice (once in the morning and once in the afternoon) during each 24-hour period. All failures shall be recorded and analyzed for cause of failure. Corrective action shall be taken when appropriate. Copies of these failure reports shall be forwarded to the contracting officer. Units for testing shall be selected from those units having successfully completed the 72 hour burn-in and the functional test (see 4.5.2.4).

4.5.2.6.2 Failure criteria. Nonconformance to 3.8 shall constitute failure of this test.

4.6 Inspection of packaging.

4.6.1 Quality conformance inspection of pack.

4.6.1.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.6.1.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.6.1.3 Examination. Samples selected in accordance with 4.6.1.2 shall be examined for the following defects. AQL shall be 2.5 percent defective.

106.	Materials, methods, and containers not as specified for level A or level B. Each incorrect material, method, or container shall be considered one defect.
107.	Boxed contents not cushioned to prevent movement or damage as specified for level A or level B.
108.	Strapping not as specified for level A.
109.	Quantities packed together exceed the size or weight limitation of the box for level A or level B.
110.	Marking missing, illegible, incorrect, or incomplete for level A, level B, or commercial.

5. PACKAGING

5.1 Preservation. Preservation shall be level A, level B, or commercial as specified (see 6.2).

5.1.1 Level A. Each demodulator shall be preserved in accordance with MIL-P-116, method IIe, and placed in an individual close-fitting box conforming to PPP-B-566, variety 1, style, type, and class optional or PPP-B-676, variety 1, style, type, and class optional.

5.1.1.1 Consolidation. Demodulators preserved as specified in 5.1.1 shall be consolidated together in a close-fitting box conforming to PPP-B-636, W6c, or

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W5c, style optional in quantities not to exceed the weight limitation of the box. The contents shall be cushioned to prevent movement or damage.

5.1.2 Level B. Preservation shall be as specified in 5.1.1 for level A, except method IA-15 shall be used in place of method IIe.

5.1.3 Commercial. Each demodulator shall be preserved in accordance with MIL-STD-1188.

5.2 Packing. Packing shall be level A, level B, or commercial as specified (see 6.2).

5.2.1 Level A. Demodulators preserved as specified in 5.1, shall be packed together in a close-fitting box conforming to PPP-B-601, overseas type, style optional, in quantities not to exceed the weight limitation of the box. Closure and strapping shall be in accordance with the appendix to the box specification.

Strapping shall conform to QQ-S-781, class 1, type I or IV, size as applicable. Unless otherwise specified (see 6.2), strapping shall be finish B. When specified (see 6.2), strapping shall be finish A.

5.2.2 Level B. Demodulators preserved as specified in 5.1, shall be packed as specified in 5.2.1 except boxes shall be domestic type. As an alternate, boxes conforming to PPP-B-636, V3c, V11c, V13, or V15c may be used when the quantity to be packed does not exceed the gross weight or size limitation of the box. Box closure and strapping shall be in accordance with the appendix to the applicable box specification.

5.2.3 Commercial. Demodulators preserved as specified in 5.1 shall be packed in accordance with MIL-STD-1188.

5.3 Marking.

5.3.1 Military. Marking for level A or level B shall be in accordance with MIL-STD-129 and the special marking specified in 5.3.3.

5.3.2 Commercial. Marking for commercial packaging shall be in accordance with MIL-STD-1188 and the special marking specified in 5.3.3.

5.3.3 Special marking. Each shipping container shall be marked with code labels in accordance with MIL-STD-129. The labels shall contain the Project Code "JID".

6. NOTES

6.1 Intended use. The demodulator is intended for use as a part of the data transmission group receiver which is a component of the Joint-Services Interior Intrusion Detection System.

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6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Time frame required for submission of preproduction model, and number of models required (see 3.2).
(c) When the Government will conduct any or all of the preproduction model examination and tests. When the Government will conduct some but not all of the preproduction examination and tests, the contracting officer should specify which examination and tests will be conducted by the Government and which examination and tests shall be conducted by the contractor (see 3.2).
(d) When Government-loaned property is not required (see 3.6).
(e) Degree of preservation and packing required (see 5.1 and 5.2).
(f) When other than finish B strapping is required (see 5.2.1).

6.3 Preproduction model. Any changes or deviations of production demodulators from the approved preproduction model during production will be subject to the approval of the contracting officer. Approval of the preproduction model will not relieve the contractor of his obligation to furnish demodulators conforming to this specification.

6.4 Government-loaned property. The contracting officer should arrange to loan the property specified in 3.6 when applicable (see 6.2).

6.5 Recycled material. It is encouraged that recycled material be used, when practical, as long as it meets the requirements of this specification (see 3.3).

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APPENDIX

FOR

CIRCUIT CARD ASSEMBLY - DEMODULATOR

10. SCOPE

10.1 Scope. The alignment and test methods contained in this appendix are to determine whether the demodulator circuit card assemblies procured under this specification conform to the requirements set forth.

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ALIGNMENT PROCEDURE FOR THE DEMODULATOR

1. PROCEDURE TO BE PERFORMED:

   (a) Preamplifier alignment.
   (b) 1270 Hz channel bandpass alignment.
   (c) 1070 Hz channel bandpass alignment.
   (d) Line fault sensitivity alignment.

2. TEST APPARATUS:

   (a) Power supply - Sorenson QRD-40-2, or equivalent.
   (b) Signal generator - Wavetack Model 110, or equivalent.
   (c) Voltmeter, AC - Hewlett-Packard 400E, or equivalent.
   (d) Frequency counter - Hewlett-Packard 5328, or equivalent.
   (e) Oscilloscope - Tektronix Model 465, or equivalent.
   (f) Attenuator - Hewlett-Packard 350D, or equivalent.
   (g) Digital multimeter - Hewlett-Packard 3469B, or equivalent.
   (h) Miscellaneous cables and interconnects.
   (i) Test fixture No. 1 - Demodulator Receiver.
   (j) Test fixture No. 2 - Demodulator Receiver.

3. ALIGNMENT PREPARATION:

   (a) Connect test equipment as shown in figure 2 using the test fixture No. 1 (see figure 1).
   (b) Adjust power supply for 20 ± 0.1 Vdc. Turn power supply off.
   (c) Initially select values of 182K ohms for R11 and 182K ohms for R17 on the demodulator.
   (d) Insert the demodulator into the test fixture and turn power supply on.
   (e) Adjust signal generator to 1170 ± 5 Hz and set the output to zero (0) dBm. With attenuator set to 0 dB, ac voltmeter should indicate 0 dB.
   (f) Connect the multimeter between TP-6 and TP-7 (gnd) of the demodulator. Voltage shall be 19.4 ± 0.1 Vdc.
   (g) Connect the multimeter between TP-4 and TP-7 (gnd) of the demodulator. Voltage shall be 6.8 ± 0.2 Vdc.
   (h) Current as indicated on meter "M" of the test fixture shall be less than 5 mA.

4. ALIGNMENT PROCEDURE:

   (a) Preamplifier alignment.

      (1) Connect the ac voltmeter between TP-5 and TP-7 of the demodulator.
      (2) Adjust the signal generator to 1270 ± 5 Hz.
      (3) Set the attenuator for -35 dB signal into the demodulator.
      (4) A signal level of -23 ± 2 dB shall be observed at TP-5. Record this level.

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(5) Tune the signal generator to 1070 ± 5 Hz. Insure signal level input to the demodulator is still at -35 dB. A signal level of -23 ± 2 db shall be observed at TP-5. If the level observed differs by more than 0.5 dB of that recorded in step (4) adjust T1 until the level observed is within 0.5 dB. Repeat steps (2), (3), (4) and (5) until level observed in step 5 is within 0.5 dB of the level recorded in step (4) without T1 adjustment.

(b) 1270 Hz channel bandpass alignment.

(1) Adjust the signal generator to 1270 ± 5 Hz and set the output to 0 dB.
(2) Connect the ac voltmeter between TP-2 and TP-7 of the demodulator.
(3) Connect the oscilloscope to TP-8 and its ground connection to J-5 of the test fixture.
(4) Set the attenuator for a loss of 20 dB.
(5) Adjust R21 on the demodulator for a peak signal on the ac voltmeter (+8 ± 1 dB). Record this reading.
NOTE: The oscilloscope will indicate a positive (+) voltage of 8.5 ± 1 Vdc.
(6) Increase the signal generator frequency until the ac voltmeter at TP-2 drops -3 dBm from the reading in step (5). The frequency at this point will be called F_H. Record this frequency.
(7) Decrease the signal generator frequency until the ac voltmeter at TP-2 drops -3dBm from the reading in step (5). The frequency at this point will be called F_L. Record this frequency.
(8) Subtract the two frequencies $F_H - F_L$. The difference is the bandwidth of the 1270 Hz channel and shall be 70 ± 10 Hz.
(9) If the bandwidth in step (8) is out of tolerance reselect R17 and repeat steps (1) through (8).

(c) 1070 Hz channel bandpass alignment.

(1) Adjust the signal generator to 1070 ± 5 Hz and set the output to 0 dB.
(2) Connect the ac voltmeter between TP-1 and TP-7.
(3) Retain the oscilloscope and attenuator as specified in paragraph b.
(4) Adjust R15 on the demodulator for a peak signal on the ac voltmeter (+8 ± 1 dB). Record this reading.
NOTE: The oscilloscope will indicate a zero (0) voltage.
(5) Increase the signal generator frequency until the ac voltmeter drops -3 dB from the reading obtained in step (4). Record this frequency F_H.
(6) Decrease the signal generator frequency until the ac voltmeter drops -3 dB from the reading obtained in step (4). Record this frequency F_L.
(7) Subtract the two frequencies $F_H - F_L$. The difference is the bandwidth of the 1070 Hz channel and shall be 70 ± 10 Hz.
(8) If the bandwidth in step (7) is out of tolerance reselect R11 and repeat steps (1) through (7).

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(d) Line fault alignment.

(1) Replace signal generator in figure 2 with test fixture No. 2 (see figure 3). Apply 20 ± 0.1 Vdc power to test fixture No. 2.
(2) Set attenuator for -28 dB signal into the demodulator.
(3) With oscilloscope connected to TP-8 on the demodulator observe a square wave with a period of 83.3 ± 1 millisecond and an amplitude of 8 ± 1 Vdc peak to peak.
(4) While observing oscilloscope increase attenuation in 1 dB steps until base band data (square wave) at TP-8 drops out. Base band dropout shall occur at -33 ± 2 dB as indicated on the attenuator. Record this level.
(5) Connect the oscilloscope to TP-3 on the demodulator.
(6) Set the attenuator to -25 dB.
(7) Adjust R35 on the demodulator to its extreme clockwise position; LED-1 shall illuminate; oscilloscope output shall go high (18 ± 1 Vdc).
(8) Increase the attentuator until its setting is within + 0.5 ± 0.5 dB of the level recorded in step 4.
(9) While observing LED-1, adjust R35 counter clockwise slowly until the LED goes out. Oscilloscope will indicate a low (0 to 1 Vdc).
(10) To check for proper setting of R35 decrease attenuator by 1 dB. LED-1 shall illuminate and TP-3 shall go high. Increase attenuator by 1 dB and observe LED-1 shall go out and TP-3 shall go low.

NOTE: Demodulator is now properly aligned and ready for functional test.

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TEST METHOD FOR THE DEMODULATOR

1. TEST TO BE PERFORMED:

(a) 1270 Hz channel center frequency and bandwidth.
(b) 1070 Hz channel center frequency and bandwidth.
(c) 1070/1270 Hz discriminator circuit sensitivity comparison check.
(d) Dynamic operational check.
(e) Voltage and current requirements.
(f) Signal sensitivity check.
(g) Line fault check.

2. TEST APPARATUS:

(a) Power supply, 2 each - Sorenson QRD-40-2, or equivalent.
(b) Signal generator - Wavetek Model 110, or equivalent.
(c) Voltmeter, AC - Hewlett-Packard 400E, or equivalent.
(d) Frequency counter - Hewlett-Packard 5328, or equivalent.
(e) Oscilloscope - Tektronix Model 465, or equivalent.
(f) Attenuator - Hewlett-Packard 350D, or equivalent.
(g) Digital multimeter - Hewlett-Packard 3469B, or equivalent.
(h) Miscellaneous cables and interconnects.
(i) Test fixture No. 1 - Demodulator Receiver.
(j) Test fixture No. 2 - Demodulator Receiver.

3. TEST PREPARATION

(a) Connect test equipment as shown in figure 2 using the test fixture No. 1 (see figure 1).
(b) Adjust power supply for 20 $\pm$ 0.1 Vdc. Turn power supply off.
(c) Insert the demodulator into the test fixture and turn power supply on.

Note: Power supply #2 is not utilized until test 4(d).

4. TEST PROCEDURE:

(a) 1270 Hz channel center frequency and bandwidth test.
(1) Tune the signal generator for an output of 1250 Hz $\pm$ 5 Hz at a level of 0 dB.
(2) Set attenuator for a loss of 20 dB. Ac voltmeter shall indicate -20 dB.
(3) Connect the ac voltmeter between TP-2 and TP-7 of the demodulator.
(4) Connect the oscilloscope to TP-8 and its ground connection to J-5 of the test fixture.
(5) Increase the output frequency of the signal generator until the signal at TP-2 peaks. Frequency at which peak reading occurs shall be 1270 $\pm$ 5 Hz. Peak reading at TP-2 shall be 8 $\pm$ 1 dB. Record this reading.
NOTE: The oscilloscope will indicate a positive (+) voltage of 8 $\pm$ 1 Vdc.

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(6) Adjust the signal generator frequency above 1270 Hz until the ac voltmeter indicates a reading 3 dB less than that recorded in step (5). Record this frequency F_H. Decrease the signal generator frequency below 1270 Hz until the ac voltmeter indicates a reading 3 dB less than that recorded in step (5). Record this frequency F_L. The 3 dB bandwidth of the 1270 Hz channel ($F_H - F_L$) shall be 70 $\pm$ 10 Hz.

(b) 1070 Hz channel center frequency and bandwidth test.

(1) Tune the signal generator for an output of 1050 $\pm$ 5 Hz at a level of 0 dB.
(2) Connect the ac voltmeter between TP-1 and TP-7 of the demodulator.
(3) Retain the remainder of the test setup as in paragraph (a).
(4) Increase the output frequency of the signal generator until the signal at TP-1 peaks. Frequency at which peak reading occurs shall be 1070 $\pm$ 5 Hz. Peak reading at TP-1 shall be 8 $\pm$ 1 dB. Record this reading.
NOTE: The oscilloscope will indicate a low voltage of 0 to 1 Vdc.
(5) Adjust the signal generator above 1070 Hz until the ac voltmeter indicates a reading 3 dB less than that recorded in step (4). Record this frequency F_H. Decrease the signal generator frequency below 1070 until the ac voltmeter indicates a reading 3 dB less than that recorded in step (4). Record this frequency F_L. The 3 dB bandwidth of the 1070 Hz channel ($F_H - F_L$) shall be 70 $\pm$ 10 Hz.

(c) 1070/1270 Hz discriminator circuity sensitivity comparison check.

(1) Subtract the smaller of the two readings, obtained in paragraphs (a) (5) and (b) (4), from the larger. The difference shall not exceed 1 dB.

(d) Dynamic operational check.

(1) Set power supply #2 to 20 $\pm$ 0.1 Vdc. Connect power supply #2 to test fixture No. 2 (see figure 3). Replace signal generator in figure 2, with test fixture No. 2. Adjust attentuator for -20 dB.
(2) Observe with the oscilloscope at TP-8 on the demodulator, a square wave with a frequency of 12 Hz and an amplitude of 8 $\pm$ 1 Vdc.
(3) Connect the frequency counter to TP-8. The frequency of the square wave shall be 12.0 $\pm$ 0.1 Hz.

(e) Voltage and current requirements test.

(1) Retain test setup as in paragraph (d).
(2) Connect the multimeter between TP-6 and TP-7 (gnd) of the demodulator. Voltage shall be 19.4 $\pm$ 0.1 Vdc.

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(3) Connect the multimeter between TP-4 and TP-7 (gnd) of the demodulator.
Voltage shall be 6.8 ± 0.2 Vdc.
(4) Current as indicated on meter "M" of the test fixture shall be less than 5 mA.
(5) Adjust main power supply in test setup to 19 ± 0.1 Vdc. and repeat test in paragraph (d). Amplitude and period of the square wave at TP-8 shall remain in tolerance. Current as indicated on meter "M" shall be less than 5 mA.
(6) Adjust main power supply to 21 ± 0.1 Vdc and repeat test in paragraph (d). Amplitude and period of the square wave at TP-8 shall remain in tolerance. Current as indicated on meter "M" shall be less than 5 mA.
(7) Adjust main power supply back to 20 ± 0.1 Vdc.

(f) Signal sensitivity check.

(1) Retain same test setup as in paragraph (d).
(2) Set attenuator for -28 dB signal into demodulator. LED-1 will be illuminated. Oscilloscope will be displaying square wave with a frequency of 12 Hz and an amplitude of 8 ± 1 Vdc peak to peak.
(3) While observing oscilloscope increase attenuator in 1 dB steps until base band data (square wave) at TP-8 drops out. Base band dropout shall occur at -33 ± 2 dB. (LED-1 shall go out). Record this reading.

(g) Line fault check.

(1) Retain same test setup as in paragraph (d).
(2) Connect oscilloscope to TP-3. Set attenuator to -28 dB. Oscilloscope shall indicate a high voltage (18 ± 1 Vdc).
(3) Increase attenuation slowly in 1 dB steps until the oscilloscope indicates a low reading. (LED-1 shall go out). Level at which TP-3 goes low shall be within 0.5 ± 5 dB of the reading obtained in paragraph (f) (3).

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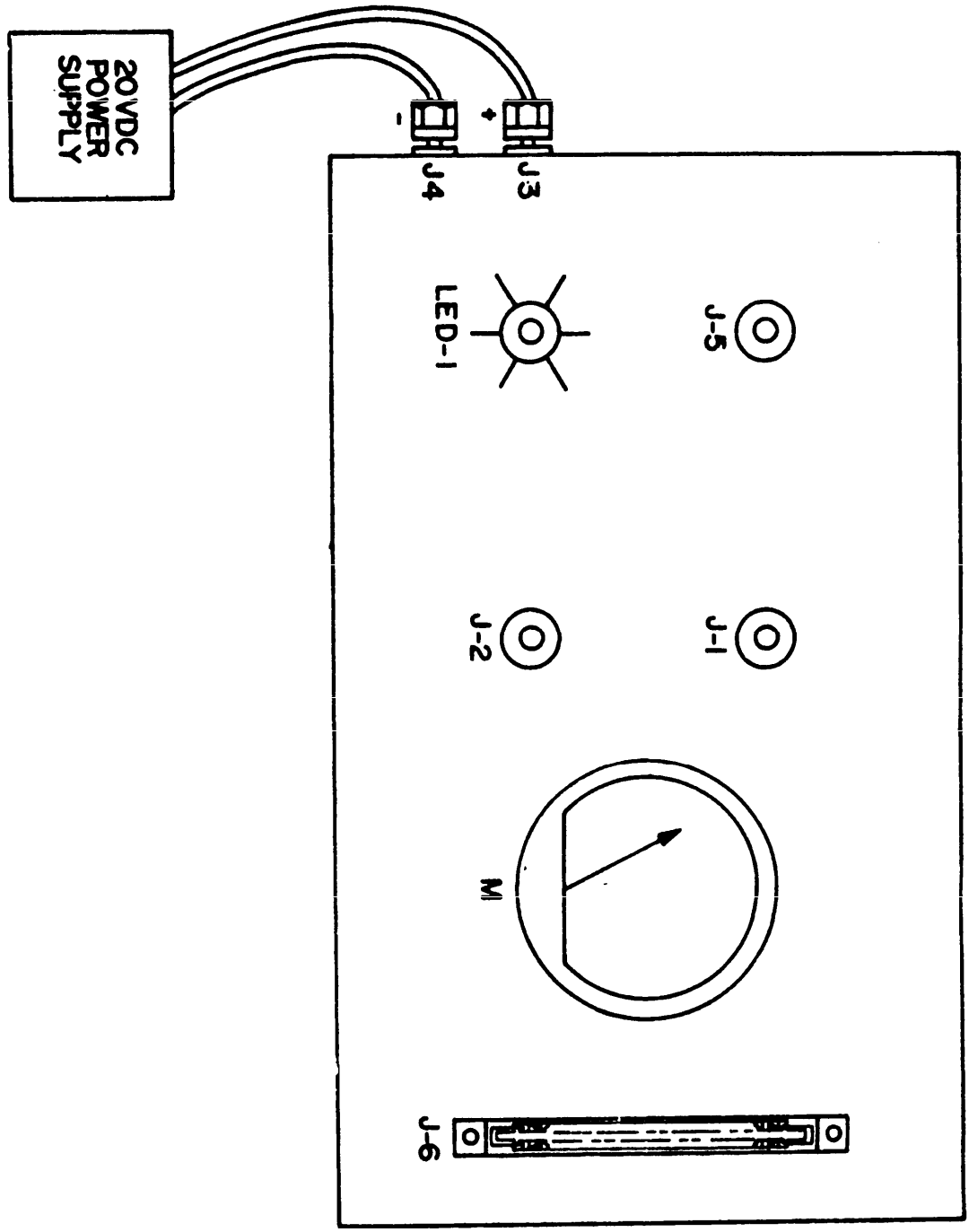
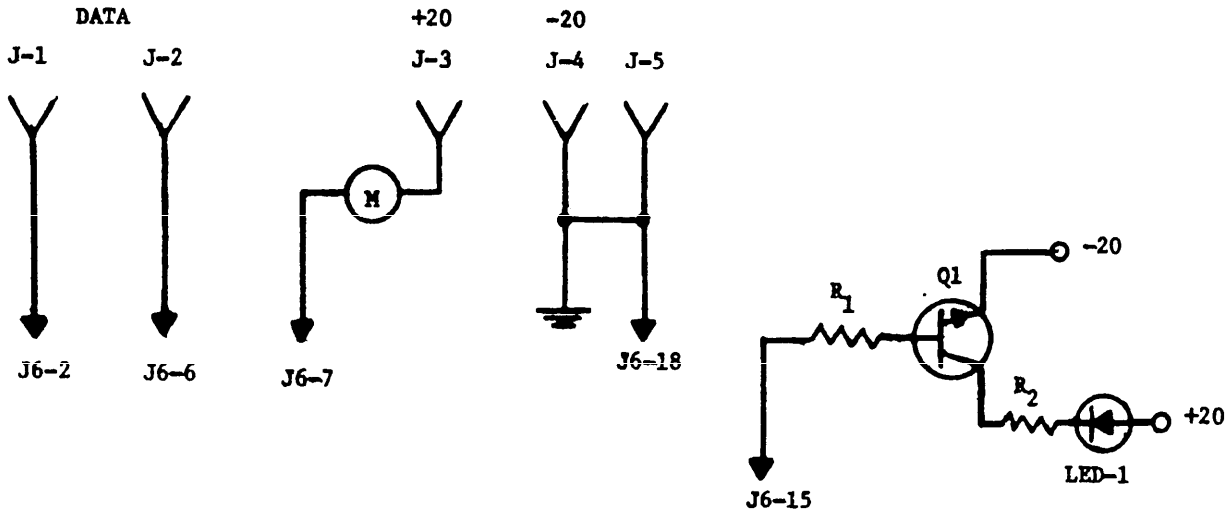


FIGURE 1. Receiver demodulator test fixture no.1.

X-3760

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Parts List

Reference Designation	Description
J-6	Connector, TRW #252-18-30-160
J-1 thru J-5	Jack, Banana
Q1	Transistor, 2N2222
R_1	Resistor, 24K, 1/2 w, 5%, carbon
R_2	Resistor, 2.2K, 1/2 w, 5%, carbon
LED-1	Diode, LED, Dialight #558-0101-001
-	Cabinet for mounting above, any convenient size
M	Meter, DC milliamp, 0-20 MA F.S.

FIGURE 1A. Test fixture no. 1 parts list

X-3761

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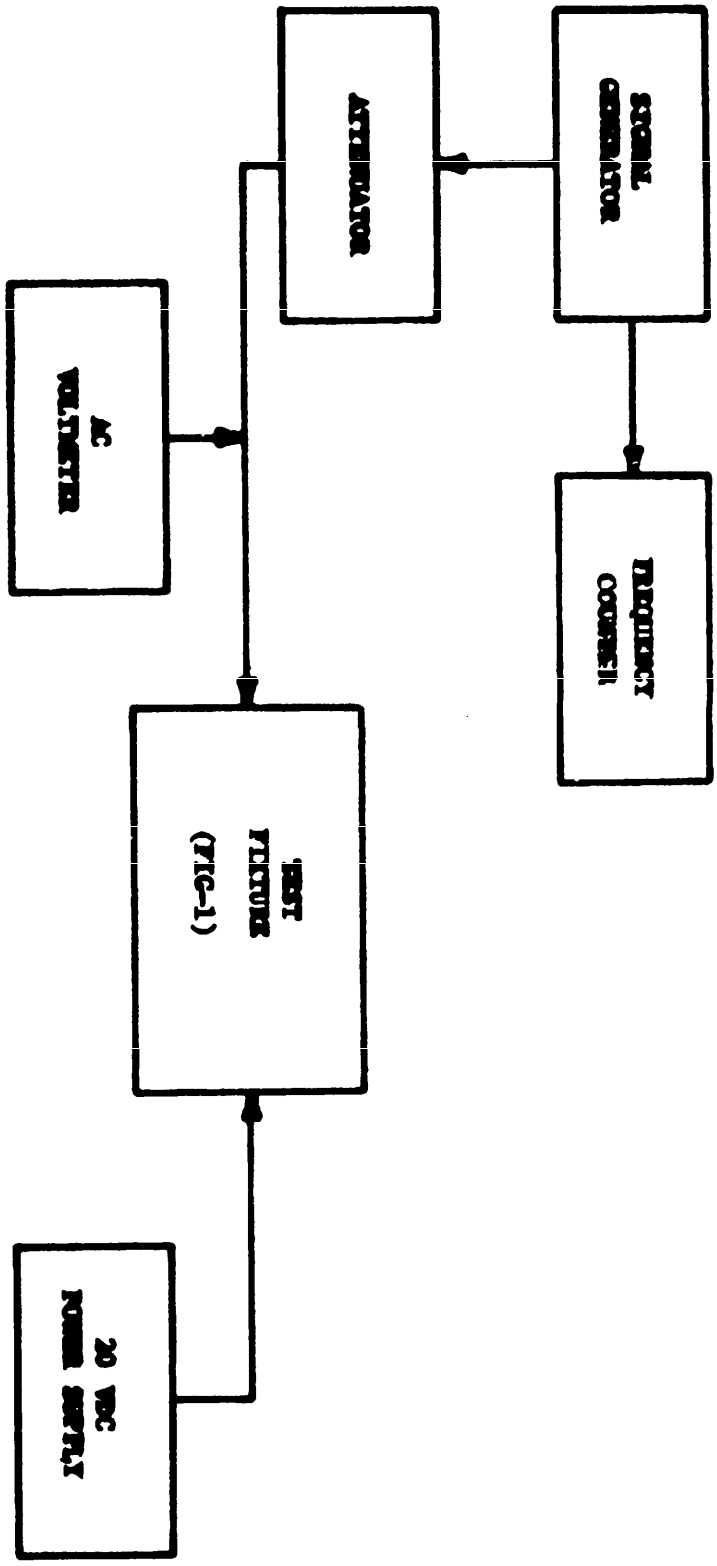
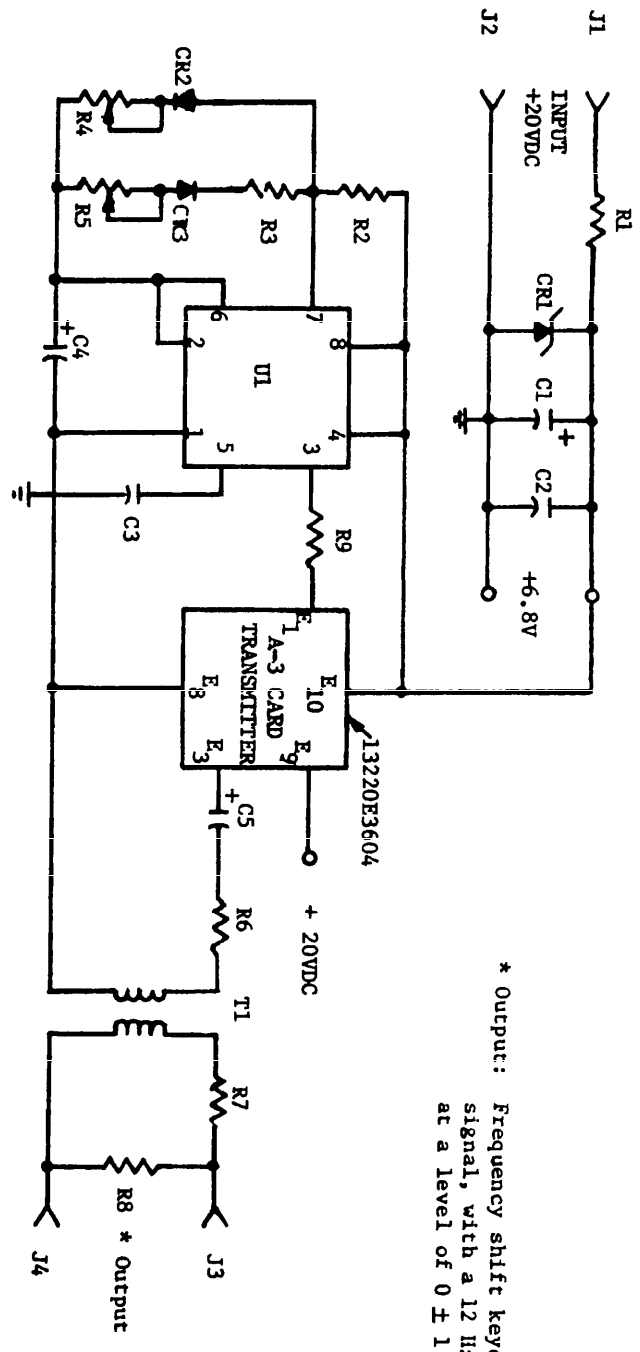


FIGURE 2. Receiver demodulator test setup.

X-3762

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* Output: Frequency shift keyed (1070/1270 Hz) signal, with a 12 Hz bit rate, and at a level of 0 ± 1 dB.

Note: 1. See Figure 3A for component values.
2. Adjust R4 for 41.66 ± 0.5 msec (positive portion only) at U1, pin 3.
3. Adjust R5 for 83.33 ± 1 msec (total period) at U1, pin 3.

FIGURE 3. Receiver demodulator fixture no.2 schematic diagram.

X-3763

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Reference Designation	Description
C1	Capacitor, Electrolytic, 47 uf ± 10%, 35 Vdc
C2	Capacitor, Disc, Ceramic, .01 uf ± 5%, 50 Vdc
C3	Capacitor, Disc, Ceramic, .01 uf ± 5%, 50 Vdc
C4	Capacitor, Tantalum, 1.0 uf ± 5%, 35 Vdc
C5	Capacitor, Electrolytic, 1.0 uf ± 5%, 15 Vdc
CR1	Diode, IN754A
CR2	Diode, IN914
CR3	Diode, IN914
R1	Resistor, 500 ohm ± 5%, 1W
R2	Resistor, 27,000 ohm ± 5%, 1/4W
R3	Resistor, 27,000 ohm ± 5%, 1/4W
R4	Resistor, Variable, 50,000 ohm ± 5%, 1/4W
R5	Resistor, Variable, 50,000 ohm ± 5%, 1/4W
R6	Resistor, 510 ohm ± 5%, 1/4W
R7	Resistor, 270 ohm ± 5%, 1/4W
R8	Resistor, 290 ohm ± 5%, 1/4W
R9	Resistor, 47,000 ohm ± 5%, 1/4W
T1	Transformer, Triad, TY304P
U1	Integrated Circuit, NE 555
Trans	A3 Card - 1322OE3604
J1 to J4	Jack, Banana
	Cabinet, suitable for mounting above components

FIGURE 3A. Test fixture no. 2 parts list.

X-3764

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